

What is missing

The missing piece is a bound on the biggest spin that does not depend on watching a run, and that stays finite for every smooth start. These tests do not have that bound.

The runs are examples. Some fade. One keeps amplifying: the matched stretch goes from a biggest spin of 60 to about 600 and is still high at time 0.35. That shows amplification can last on the times tested. It does not say what every smooth start does.

The grids do not agree on the size of the peak. At time 0.12 it is 194 on 48 cubed, 200 on 64 cubed, and 234 on 80 cubed. The hot region is still many cells wide, so it is not a one-cell spike. The peak getting larger on a finer grid means that climb has not settled. A finer grid has not yet been run long enough to see if it turns down.

Smoothness on an interval follows if the time integral of the biggest spin stays finite. On the matched stretch that integral is 120 through time 0.36. It was still rising. A finite number on one interval is not a guarantee for later times, and it is not a guarantee for every start.

Smoothing, pressure, and the spin ratio do not close it. Smoothing was smaller than the steepening at the fastest point, and the speed still faded on the pulled tube. Pressure changed sign. On the matched stretch the stretch stayed ahead often enough for the biggest spin to keep climbing. None of those is a brake that has been shown to win for every start.

The cutoff choice changes the picture. A cutoff that moves with the peak makes the hot region look smaller: 27,191 cells, then 485. A fixed cutoff does the opposite: above 50 the count goes 930, then 15,815, then 40,524. The core tightens and more of the cube crosses a fixed spin at the same time. That is not a collapse to a point, and it is not a settled spread either.

The honest claim is a tested margin on selected starts over finite times. The claim that is missing is the one that covers every smooth start for all later time.